# UNIT - III

## TQM PRINCIPLES

Customer Satisfaction - Employee Involvement - Continuous Improvement - Supplier Partnership - Performance Measures

### 1. Customer Satisfaction

> The customer is the **king** - the one who receives the product or service. Customer satisfaction = the feeling that the product/service has met or exceeded the expectation.

- **Satisfaction = Perception of performance - Expectation.** If P > E the customer is delighted; if P < E dissatisfied.
- Types: **External customer** (buys/uses the product) and **Internal customer** (next person/department in the process).
- Customer is the one who defines quality - "voice of the customer".
- A satisfied customer tells 3-5 people; a dissatisfied customer tells 10-20 people. It costs 5 times more to get a new customer than to keep an old one.

#### Customer perception of quality

- Quality is **what the customer perceives,** not what the producer thinks.
- Factors that shape perception: performance, features, service, warranty, brand name and reputation, price, advertisement, past experience, word of mouth.
- Perceived quality is affected by: the product, **the service** that goes with it (delivery, installation, help desk), and the **organisation's image.**
- Needs are of 3 kinds: **Basic** (must be there), **Performance** (more is

better) and **Excitement/delighters** (unexpected features) - the **Kano model.**




### Kano model (note)

- Basic needs: absence causes dissatisfaction, presence does not satisfy (brakes in a car).
- Performance needs: satisfaction rises with level (mileage).
- Delighters: unexpected - big satisfaction (e.g., free pick-up service).

### Ways to understand customers (feedback tools)

- Surveys and questionnaires, focus groups, comment cards, toll free numbers, customer visits, report cards, the internet/social media, mystery shoppers, sales and service staff, warranty data, complaint analysis.
- Customer Satisfaction Index (CSI) - tracking over time; comparing with competitors.

## 2. Customer Complaints

A complaint is an expression of dissatisfaction. Complaints are **free consulting** - opportunities to improve and to retain the customer.

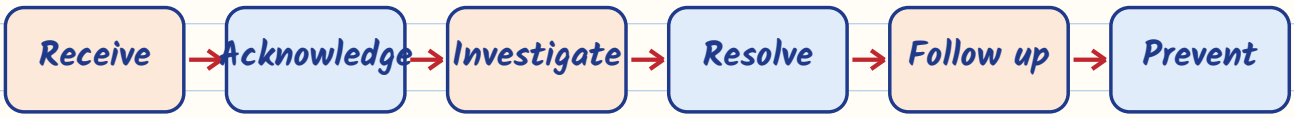
- Only about 4% of unhappy customers complain; the rest leave silently ("iceberg").

### Steps to handle complaints effectively

1. **Listen** actively; let the customer finish; do not argue.
2. **Apologise and show empathy;** take responsibility.
3. **Record** the details - what, when, who, where.
4. **Solve** the problem quickly - replace, repair, refund; empower front

line staff.

5. **Follow up** to confirm satisfaction.
6. **Analyse** causes, take corrective action and feed back to design/production.




### Best practices

- Make it easy to complain (toll free number, email, app).
- Respond within a fixed time (acknowledge in 24 hours).
- Maintain a complaint database; use Pareto analysis on causes.
- Train employees; reward complaint handling.
- "Service recovery paradox": a well handled complaint can create a more loyal customer than before.

### Complaints and the quality system

- ISO 9001 requires monitoring of customer perception and records of complaints and corrective action.

## 3. Service Quality

- Services are **intangible, inseparable** (produced and consumed together), **heterogeneous** (vary), **perishable** (cannot be stored).
- Hence quality must be **built into the process and the people,** because there is no chance to inspect before delivery.

### SERVQUAL gaps model (Parasuraman, Zeithaml, Berry)

- **Gap 1:** customer expectation vs management perception.
- **Gap 2:** management perception vs service quality specification.
- **Gap 3:** specification vs service delivery.
- **Gap 4:** service delivery vs external communication (promises).
- **Gap 5:** expected service vs perceived service (the customer gap) =

result of gaps 1 to 4.

### Dimensions (RATER)

- **Reliability, Assurance, Tangibles, Empathy, Responsiveness.**

### Translating service quality into practice

- Service blueprinting, standards for response time, moment of truth (every contact), training front-line staff, service guarantee, use of technology (ATM, e-banking), measure through surveys, mystery shopping.
- **Moments of truth** (Jan Carlzon, SAS): each customer-employee contact shapes perception.

### Service quality vs product quality

| Product | Service |
|---|---|
| Tangible | Intangible |
| Can be inspected before delivery | Quality seen during delivery |
| Stored | Cannot be stored |
| Produced first, consumed later | Produced and consumed simultaneously |

## 4. Customer Retention

> Customer retention = keeping existing customers satisfied so that they continue to buy. It is the **result of customer satisfaction.**

- Retaining is cheaper than acquiring; loyal customers buy more, pay more, refer others.
- Retention rate = (Customers at end - New customers) / Customers at start x 100.
- **Customer loyalty** is stronger than satisfaction - "completely satisfied" customers are far more loyal than "satisfied" ones.

### Factors for retention

- Product quality and consistency, on-time delivery, after-sales service, fair price, quick complaint redress, trust, communication.

### Retention strategies

- Loyalty / reward programmes, personalised service, CRM software, customer clubs, warranty and AMC, relationship marketing, customer feedback loop and follow up, delight them beyond expectation.
- Customer lifetime value (CLV) thinking.

Remember: Satisfaction -> Loyalty -> Retention -> Profit.

## 5. Employee Involvement

Employee involvement means that **all employees participate** in decision-making and improvement; they feel ownership of the quality of their work.

- Employees closest to the job know most about the problems.
- Benefits: better ideas, commitment, reduced resistance to change, higher morale, lower turnover.
- Means: suggestion scheme, quality circles, teams, empowerment, job enrichment, training, recognition.

### Motivation

- **Motivation** is the internal drive to act towards a goal. In TQM, managers create conditions in which employees want to do quality work.

### Maslow's hierarchy of needs




- Lower needs satisfied -> higher needs drive behaviour. Job security,

belonging in teams, recognition and growth are important in TQM.

### Herzberg's two factor theory

| Hygiene factors (dissatisfiers) | Motivators (satisfiers) |
|---|---|
| Salary, working conditions, company policy, supervision, job security | Achievement, recognition, responsibility, the work itself, advancement, growth |

- Absence of hygiene factors makes people unhappy; but only motivators can produce satisfaction.

### Other theories

- **McGregor's Theory X and Y:** X - people dislike work, need control; Y - people seek responsibility, self-direction. TQM follows Theory Y.
- **Vroom's expectancy theory, McClelland's needs** (achievement, power, affiliation), **Alderfer's ERG.**

### How to motivate for quality

- Clear goals, feedback, involvement, fair pay, recognition and rewards, training, job rotation and enrichment, trust and open communication, drive out fear.

### Recognition and reward

- Recognition = acknowledgement of effort (thank-you, certificate, notice board); Reward = tangible (bonus, gift, promotion).
- Should be timely, specific, fair, linked to team performance; do not replace with annual merit rating (Deming).

## 6. Empowerment and Teams

**Empowerment** is investing people with authority to make decisions and take action in their own area of work, with responsibility and accountability.

## Conditions for empowerment

- Everyone understands the need, knows the system and the goal; has training and skills; has information, tools and resources; boundaries are clear; management support, trust; rewards linked.
- Steps: create trust, share information, give training, delegate authority gradually, give feedback.
- Benefits: faster decisions, ownership, motivation, quality at source.

## Teams

> A team is a group of people with complementary skills committed to a common purpose, performance goals and approach for which they hold themselves mutually accountable.

- Types of teams: **Process improvement teams, cross-functional teams, natural work teams, self-directed (self-managed) teams, quality circles** (voluntary, 6-12 members, meet regularly), task force / ad-hoc teams, **suggestion** based.

## Stages of team development (Tuckman)




## Characteristics of successful teams

- Clear mission and goals; open communication; defined roles; trained leader and members; use of problem solving tools; mutual respect and trust; management support; recognition.

## Quality circles

- Started in Japan 1962 (Ishikawa/JUSE). Voluntary small group of the same work area; meets 1 hour weekly; **facilitator,** leader, members; identifies, analyses and solves problems; presents to management; seven QC tools.

### Problem solving in teams

- Brainstorming, nominal group technique, 5 Why, fishbone, Pareto, PDCA.

### Team meeting ground rules

- Agenda, time keeping, minutes, one person at a time, decisions by consensus.

## 7. Continuous Process Improvement

> Continuous improvement **(Kaizen)** means improving every process, every day, in small steps - with no end.

- Process: a set of activities that converts input into output. Every process has an owner, suppliers and customers; variations can be reduced and wastes removed.
- Basis: **input -> process -> output,** with feedback.

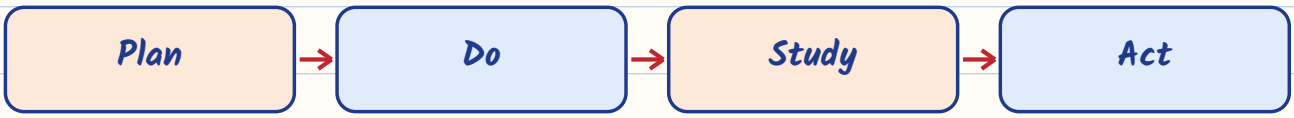
### Juran Trilogy




- **Quality planning:** identify customers and needs, design product and process; hand over to operations.
- **Quality control:** operate process, compare actual with goal, act on the difference (keep chronic level and prevent sporadic spikes).
- **Quality improvement:** break through to a new, lower level of chronic waste - project by project.
- Juran diagram: cost of poor quality vs time - planning sets level; sporadic spike is fixed by control; improvement lowers chronic waste.
- Juran's 10 steps to improvement: awareness, goals, organise, train, projects, report progress, recognise, communicate results, keep score,

maintain momentum.

## PDSA Cycle (Shewhart/Deming)




- **Plan:** identify the opportunity and plan the change; analyse the present process, collect data, set targets.
- **Do:** implement the change on small scale (pilot).
- **Study (Check):** analyse the results; what did we learn?
- **Act:** if successful standardise and institutionalise; else repeat with new knowledge.
- Repeat continuously - the improvement spiral. A better name than PDCA because "study" stresses learning.
- **Process improvement 7-step:** define the process -> Select target -> Analyse data/causes -> Develop solution -> Implement -> Verify results -> Standardise.

## Kaizen

**Kaizen** = "Kai" (change) + "Zen" (for the better). Masaaki Imai - continuing improvement involving everyone: top managers, managers and workers.

- Kaizen is **people oriented,** small steps, low cost, bottom up as well as top down, process based, and long term. Versus **innovation:** radical, big investment, technology led.
- Kaizen tools: **5S** (Seiri-sort, Seiton-set in order, Seiso-shine, Seiketsu-standardise, Shitsuke-sustain), suggestion system, **Gemba** (go to the actual place), **muda** elimination (7 wastes: overproduction, waiting, transport, over-processing, inventory, motion, defects), standard work, PDCA/SDCA.

- Kaizen events (blitz) - 3-5 day team projects.

| Kaizen | Innovation |
|---|---|
| Small, continuous | Large, one-time |
| Everyone | Few specialists |
| Low investment | High investment |
| Process focus | Result focus |

## 8. Supplier Partnership

> Quality of final product depends on bought-out materials (often 50-80% of cost). Suppliers must be treated as **partners** rather than adversaries.

### Partnering

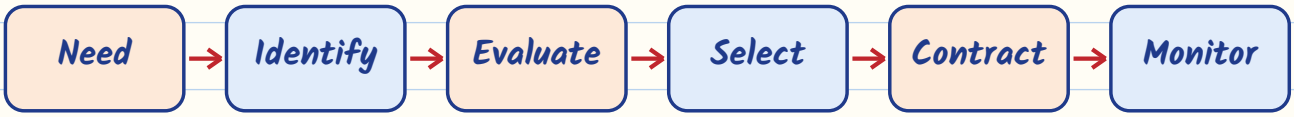
- Long term relationship based on mutual trust, shared goals, shared information and risk; win-win.
- Deming: end the practice of awarding business on price alone; use **single-source** where possible.
- Benefits: lower cost, better quality, JIT delivery, less inspection, joint development, faster innovation.
- Requirements: top management support, mutual training, clear specs, communication, share of improvement gains.

### Sourcing

- **Sole sourcing** (one supplier) - good relation, lower cost, but risk; **Multiple sourcing** - risk shared, competition, but less commitment; **Single/dual sourcing** is a compromise.
- Make or buy decision; local vs global; certified suppliers (ISO 9001) require less inspection.

## Supplier selection (Besterfield - 9 steps)

1. Determine the need, specification and quantity.
2. Identify potential suppliers (catalogues, trade journals, past records).
3. Short list; send questionnaire/request for quotation.
4. **Evaluate:** financial status, quality system, capacity, technology, location, delivery, service.
5. Visit and audit; sample evaluation.
6. Negotiate the contract; select supplier.
7. Place order; monitor; feedback.
8. Review performance regularly.
9. Develop the supplier.




## Supplier rating

- Rating = a numerical score of supplier performance used for ranking, rewarding and decision on future orders.
- Criteria: **Quality** (percentage accepted, rejection ppm), **Delivery** (on time %), **Price/cost, Service/response,** quality system certification.
- Eg. Weighted rating: Quality 40 + Delivery 30 + Price 20 + Service 10 = 100 points.
- Grades: A (preferred) 90+, B (approved) 75-89, C (conditional) 60-74, D (reject) below 60.
- Used for: certification, **vendor of the year** award, reducing inspection of A-class suppliers, corrective action for weak ones.

## Relationship development

- Stages: transactional -> cooperative -> partnership -> strategic alliance.

- Actions: supplier conferences, joint training, early supplier involvement in design, long-term contracts, share forecasts, supplier councils, recognition, joint improvement projects, resident engineers.
- Supplier certification (e.g., Ford Q1, ISO 9001, QS 9000).

## 9. Performance Measures

### Basic concepts

> Performance measure = a quantified way of knowing how well the processes and the organisation are performing against goals. "You cannot control what you cannot measure."

- Purposes: identify problems, track progress, drive improvement, motivate, help in decisions, benchmarking.
- Good measures are: **simple, relevant, understandable, timely, accurate, linked to goals, few in number, owned by someone.**
- Measure the right things: customer, process, employee, supplier, financial - not only financial (lagging) but leading indicators.
- Types: **Output** (results) and **process** (leading) measures; Management measures: customer satisfaction, defects ppm, cycle time, on time delivery, cost of quality, absenteeism, training hours, supplier rating.

### Strategy for performance measures

- Link to the organisation's strategy; develop measures at each level; use **Balanced Scorecard** - financial, customer, internal business process, learning and growth.
- Steps: identify the process -> define the measure -> set target/standard -> collect data -> analyse -> compare -> take action -> review measure.
- **Quality rating index, Customer satisfaction index, Process capability, Six sigma level, First pass yield, ppm, Return on**

quality.

- Display: control charts, trend charts, dashboards, "scoreboards" displayed where workers can see.

| Perspective | Sample measure |
|---|---|
| Financial | ROI, cost of quality |
| Customer | CSI, complaints, retention |
| Internal process | Defects ppm, cycle time, yield |
| Learning and growth | Training hours, suggestions per employee |

### Measures for service organisations

- Waiting time, errors per transaction, call abandonment rate, number of complaints, accuracy.

### Performance measure case studies

- **Case: Motorola** - used defects per million; cycle time; led to Six Sigma.
- **Case: A bank branch** - measured waiting time, errors, and customer satisfaction; found that long queue at 12-2 pm; added counter and reduced wait from 12 to 5 minutes; satisfaction index up 15%.
- **Case: Xerox** - customer satisfaction, benchmarking market share; improved supplier rating and reduced the number of suppliers from 5000 to 500.
- **Case: Wipro/Infosys** - use balanced scorecard and Six sigma metrics.

## Important Exam Questions (Unit III)

### Short answer

- Define customer perception of quality.
- Write short notes on the Kano model / the 5 service gaps.
- What is empowerment? What is Kaizen?
- Mention the steps in the PDSA cycle.

- What is supplier rating?

Long answer

1. Explain the customer satisfaction model and the methods of handling complaints.
2. Discuss Maslow and Herzberg's theories and their use in motivating employees.
3. Explain the Juran trilogy with a neat sketch.
4. Explain the PDSA cycle and Kaizen concept with examples.
5. Describe the steps in supplier selection and rating, and the relationship development process.
6. Discuss performance measures - basic concepts and strategy.